Lecture-5
Fluid Statics III: Forces on Bodies

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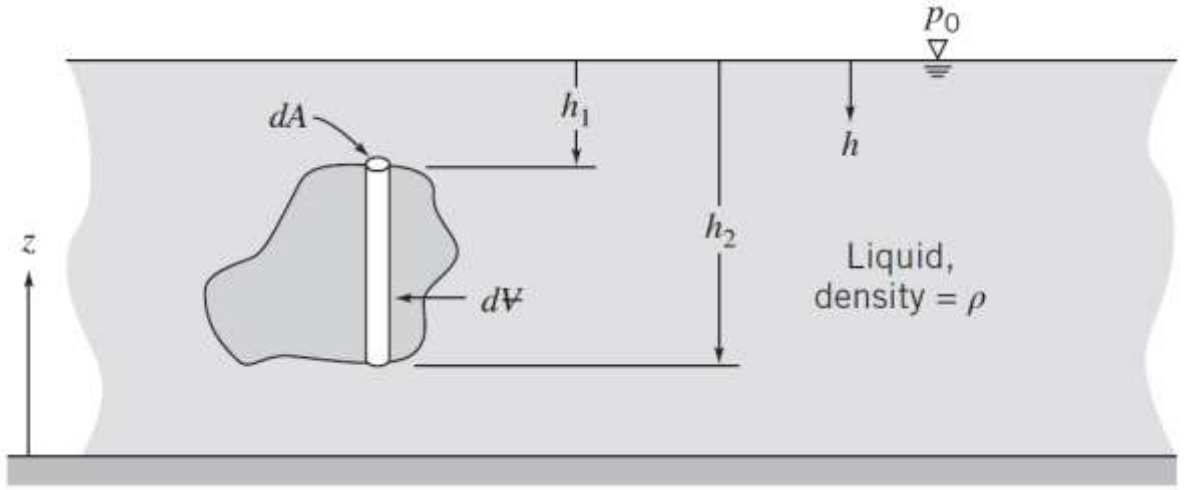
Learning Objectives

By the end of this lecture, you should be able to

1. Describe the physical origin of the **buoyant force**
2. Distinguish between the **center of buoyancy** and the **center of gravity**
3. Predict whether a floating body is in a state of **stable, neutral, or unstable equilibrium**

Force on a submerged body

- An object **immersed** in a fluid **or floating** on its surface experiences a **net upward force** due to the fluid pressure. This force is called the **buoyant force**.



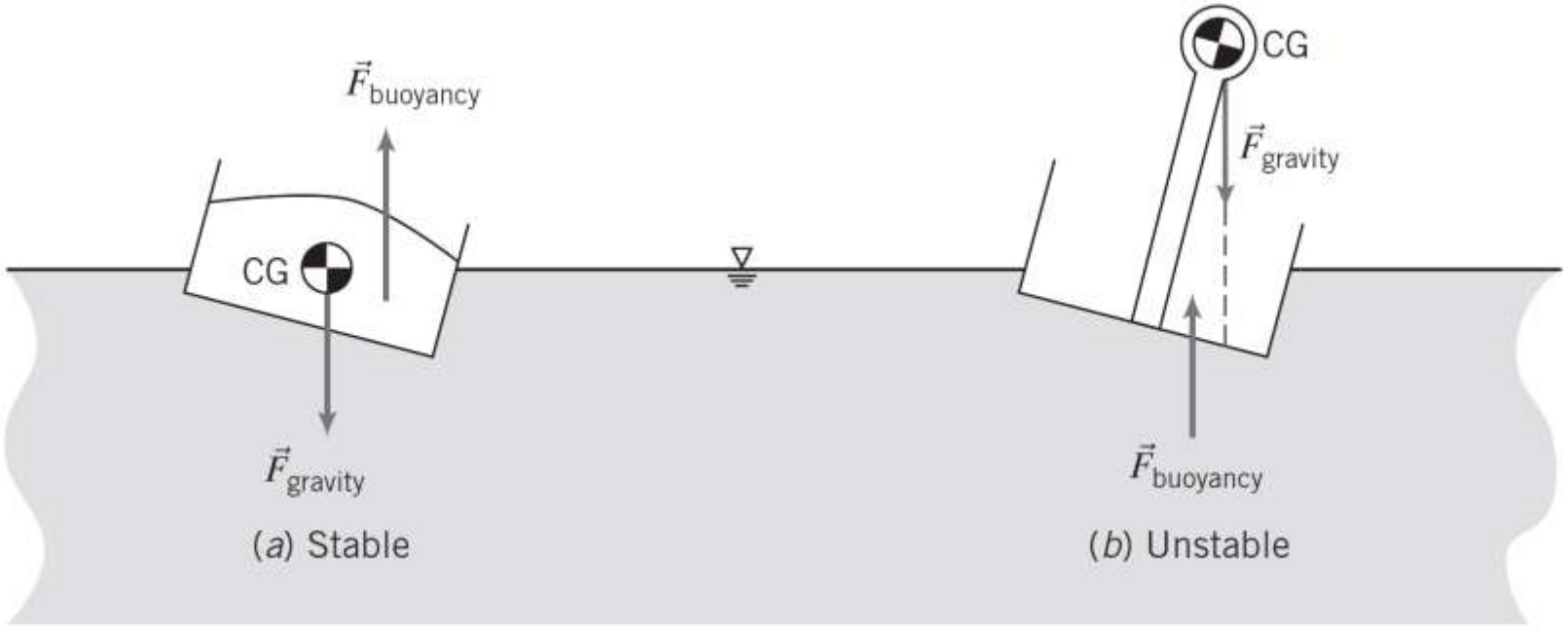
$$F_{\text{buoyancy}} = \rho g \forall$$

- Some texts state the Archimedes principle as **"Buoyant force is equal to the weight of fluid displaced"**

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Stability of a submerged body

- Assuming a body has no net force acting on it, stability examines the capability of the body to restore its original configuration when subjected to **an angular displacement**.
- Since floating bodies are in equilibrium under the action of gravity (body) and buoyant forces, their **line of action** determines the stability of the body.



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Fluids in rigid body motion

- The **fundamental equation of fluid statics can be revised** when the fluid is undergoing a rigid-body motion.

$$-\nabla p + \rho \vec{g} = \rho \vec{a}$$

$$\left.\begin{aligned} -\frac{\partial p}{\partial x} + \rho g_x &= \rho a_x \quad x \text{ direction} \\ -\frac{\partial p}{\partial y} + \rho g_y &= \rho a_y \quad y \text{ direction} \\ -\frac{\partial p}{\partial z} + \rho g_z &= \rho a_z \quad z \text{ direction} \end{aligned}\right\}$$

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Revisiting the learning objectives

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